Twitch network analysis

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Abstract

This research investigates the structural impact of Twitch Teams on audience diffusion and digital social mobility. By constructing a 10,000-node network of streamers, we analyze the platform's topology through link prediction heuristics and stochastic cascade modeling. Our findings reveal a highly assortative "Small World" network characterized by rigid compartmentalization and a structural "moat" around elite hubs. Link prediction metrics confirm that team formation is driven by local triadic closure, reinforcing existing communities rather than bridging distant social strata. Furthermore, Monte Carlo simulations identify a delayed, two-phase diffusion dynamic, demonstrating that raid-driven influence remains trapped within elite-adjacent clusters, with peripheral growth occurring only through rare, stochastic bridging ties. Ultimately, our analysis concludes that Twitch Teams function as proximity multipliers that reinforce the platform's status quo, effectively decoupling growth opportunities from the "long tail" of emerging creators and stifling platform-wide social mobility.

1 Data collection

The primary objective of this project phase is to autonomously identify, collect, and build a large-scale network dataset from an online source. Twitch, a leading live streaming platform, was selected as the data source due to its rich, interconnected community of content creators. By querying the official Twitch API via OAuth2 client credentials, we constructed a complex network that models how streamers are interconnected within the platform's ecosystem.

To formally define the network topology, we formally defined the network topology and we specified the following entities and relationships:

- **Nodes (Vertices):** Each node represents a unique Twitch streamer.
- **Edges (Links):** A directed edge is established between a source streamer and a target streamer if they share a connection through a "Twitch Team". Teams are groups created by streamers to unite under a single banner, making co-membership a strong indicator of a social or professional relationship.

To build the network from scratch, a Breadth-First Search (BFS) algorithm was implemented in Python. The crawler was initialized with a queue of highly popular Italian streamers (e.g., *tumblurr, pow3r, grenbaud*) acting as "seeds". For each user in the queue, the script queried the API to identify the Twitch Teams they belong to. Subsequently, it extracted all other members within those teams, creating an edge between the source and each discovered member. Newly discovered, unvisited nodes were then appended to the queue. This iterative process continued until the set threshold of 10,000 unique nodes was successfully reached.

1.1 Attribute selection

Once the base topology (nodes and edgelist) was established, secondary API calls were orchestrated to fetch specific node attributes. Due to the scale of the dataset, this was handled efficiently via batching (e.g., retrieving up to 100 users per request). The collected attributes include:

- **Display Name (Label)**
- **Total Views**
- **Total Followers**
- **Broadcaster Type** (e.g., partner, affiliate, or normal)

Twitch categorizes streamers into three tiers: Base streamers who lack monetization tools, Affiliates who unlock revenue features upon reaching growth milestones, and Partners who receive premium benefits like prioritized support and guaranteed video quality. While entry-level creators rely on third-party services, Affiliates and Partners gain access to native monetization tools with increasingly higher levels of support and customization as they progress.

The most significant challenge during data collection was the platform's API rate limiting. Twitch enforces a strict limit (approximately 800 requests per minute). Given that fetching followers requires individualized queries (resulting in at least 10,000 distinct requests), an automated safeguard was engineered. The script actively monitors for HTTP 429 (Too Many Requests) responses, dynamically parses the *Retry-After* header, and pauses execution for the exact duration required before safely resuming.

1.2 final network

As per the project requirements, the final network consists of over 10,000 nodes. All scripts used to query the API, handle the crawling logic, and enrich the dataset have been uploaded to the group's dedicated GitHub repository. The finalized datasets, including the edgelist and the enriched nodelist, have been compressed and made available in the repository's root directory for full reproducibility.

ID	Label	Broadcaster Type	Total Followers
77827128	Tumblurr	partner	2006052
539141014	aNcMedia	partner	4582
179279586	Kertus	partner	170260
212737350	NinjoTv	partner	112211
20844000	JodyJDC	partner	163204

Table 1: first 5 nodes overview

2 Network analysis

In this section, a graph will be constructed based on the data downloaded earlier, and an overview will be provided in order to better

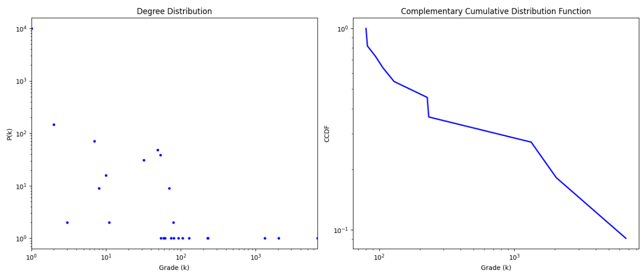

Figure 1: degree distribution

understand the nature of the dataset. For simplicity, will be analyzed the undirected graph.
The number of nodes is 10356, while the number of edges is 14192. The arithmetic mean degree is 2.74, which is not really representative of a typical node; the degree distribution is extremely right-skewed, with the median and mode both equal to 1 and 96% of nodes having degree exactly one.

The graph showing the degree distribution helps illustrate its asymmetric nature: the first panel displays the degree distribution on a log-log scale. The trend, which spans several orders of magnitude, is an indicator of a heavy-tailed distribution—a fundamental characteristic of scale-free networks, where the vast majority of vertices have very low degrees, while a very small number of "hubs" have a massive number of edges, peaking at over 6,000 connections. To evaluate the power-law behavior more rigorously, and thus reduce the statistical "noise" generated at the tail of the distribution due to finite-sample-size effects, the Complementary Cumulative Distribution Function (CCDF) was plotted. The CCDF represents the probability that a randomly selected node has a degree strictly greater than a value k. The straight line confirms that the degree distribution decays according to a power law of the form: $P(k) \sim k^{-\alpha}$.

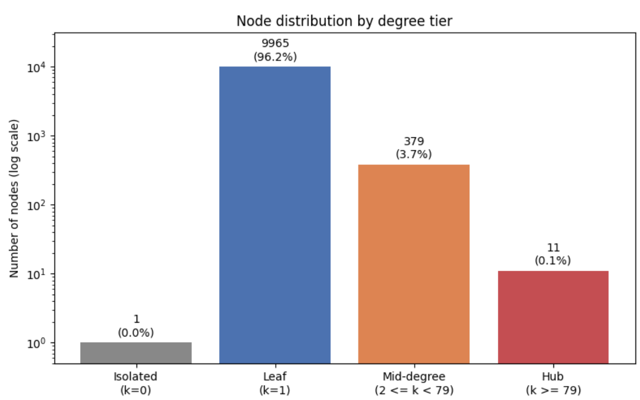

Figure 2: node distribution by degree tier

The bar chart in shows even more clearly how the degree distribution of the nodes is concentrated around 1, while there are very small hubs with thousands of nodes.

The network has a density of $d \approx 0.00026$, indicating extreme sparsity, which, however, does not result in a disconnected network. In fact, despite this sparsity, evaluating the connected components, reveals that 10355 out of 10356 belong to a single Giant Connected Component, leaving only one isolated node.
The second micro-component of the network is a single streamer, which is therefore not connected to any other streamer. This can be due to two main causes:

- The streamer in question has no actual connections to any other streamers, so he is represented as an isolated node in the network.
- The sample includes approximately 10,000 nodes, so some streamers—which would actually have connections to the isolated streamer may not be present in the dataset

It has been decided to treat the isolated streamer as an independent streamer; therefore, subsequent analyses will be carried out excluding it from the dataset

Table 2: Comparison between the real network (RN) and the synthetic null models (ER, BA), matched on N and average degree.

	RN	ER	BA
Nodes (N)	10356	10356	10356
Edges (E)	14192	14285	14098
Avg. degree	2.74	2.76	2.72
Max degree	6680	12	237
Density	2.65×10^{-4}	2.66×10^{-4}	2.63×10^{-4}
Avg. clustering	0.0244	0.0005	0.0018
# Components	2	717	1
Giant comp. (%)	99.99	92.26	100.00
Avg. path length*	3.11	9.03	5.96
Power-law α	1.52	2.25	2.70

2.1 Comparison with Erdős–Rényi

The Erdős–Rényi model was generated with the same number of nodes and an edge probability equal to the empirical density ($p \approx 2.65 \times 10^{-4}$), producing a network with an almost identical average degree. Despite this parity, the two networks differ radically in structure, The ER degree distribution is well approximated by a Poisson centered on the mean: the maximum degree observed is 12, against 6680 in the real network.
This highlights the extreme heterogeneity of the empirical graph, which is found to be incompatible with a purely random process.

2.2 Comparison with Barabási–Albert

The Barabási–Albert was built using dual-attachment variant (mixing $m_1 = 1$ and $m_2 = 2$ edges per new node with probability 0.63) in order to reproduce the average degree of 2.74.
Because BA connects new nodes preferentially to already well-connected ones, it produces a heavy-tailed degree distribution and, by construction, a fully connected giant component.
However, even this mechanism falls short of reproducing the extremity of the real hubs: the maximum degree in BA reaches only 237, and the fitted exponent is markedly steeper ($\alpha_{BA} = 2.70$ against the $\alpha_{RN} = 1.52$).

The real network is, thus, more hub-dominated than pure linear preferential attachment would predict. A plausible explanation is that popularity on Twitch is not only a "rich-get-richer" process driven by peer mentions, but is also reinforced by external factors (platform algorithms, partner status, pre-existing off-platform fame) that let top streamers accrue connections faster than linear preferential attachment alone implies.

As with ER, clustering remains the main structural gap: BA produces almost no triangles (0.0018), since new nodes attach to hubs but not to each other, while the real network's clustering (0.0244) reflects genuine local community structure among peer streamers. This is arguably the clearest evidence that neither null model, despite reproducing degree heterogeneity or connectivity individually, captures the combination of hub dominance.

3 Community Discovery

The modular structure of the Twitch network was extracted using three different algorithms, provided by the library `CDlib` [?]: **Leiden**, **Infomap**, and **ANGEL**.
Each algorithm targets a specific property of the following network. **Leiden** is used as the disjoint-partition baseline on the undirected projection of the graph. Infomap, conversely, is applied directly to the directed graph. Since edge direction in this dataset reflects the order in which Team co-membership was discovered during crawling rather than a genuinely asymmetric social relation (Section 1), this comparison does not test follow-style asymmetry; instead, it serves as a robustness check, quantifying how sensitive the detected community structure is to a directionality artifact introduced purely by the data collection methodology. **ANGEL** is used to detect overlapping communities, for testing whether streamers are better described as belonging to a single audience niche or to several simultaneously, a dynamic that disjoint-partition methods cannot express by construction.

Table 3: Community size statistics.

Algorithm	#Com.	Mean size	Med. size	Max size
Leiden	8	1,294	257	6,547
Infomap (dir.)	10	1,036	133	6,547
ANGEL (overl.)	6	45.5	41.5	96

The gap between mean and median size for Leiden and Infomap signals a heavily skewed size distribution, as noted above: both algorithms produce one giant community of comparable size (≈6,500 nodes) plus a long tail of small ones; while ANGEL's communities are uniformly small, reflecting its restriction to dense local cores rather than a full partition of the graph.

The figure illustrates how Leiden and Infomap overlap for the three largest communities, with Infomap being divided into smaller sections than Leiden. ANGEL remains consistently smaller and covers a narrower range, as confirmed by its low coverage

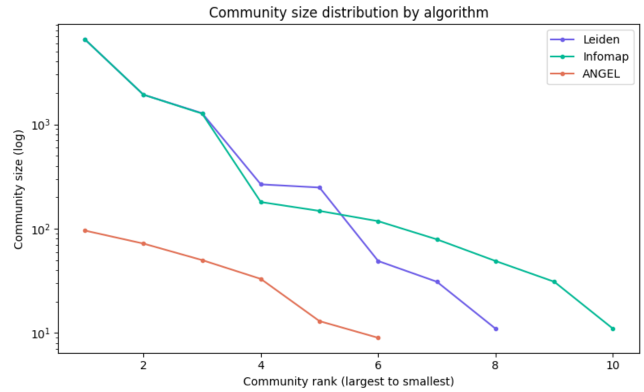

Figure 3: community size distribution

3.1 Evaluation

A Newman-Girvan modularity score has been reported for both Leiden and Infomap in order to compare on equal footing a modularity-like score and a random-walk description length; for ANGEL just AID and IED has been used (modularity is not well defined for overlapping partitions, since its null model assumes each node belongs to exactly one community).

Table 4: Community evaluation metrics.

Algorithm	Coverage	Modularity	AID	IED
Leiden	100%	0.699	13.55	0.33
Infomap (dir.)	100%	0.261	13.88	0.31
Infomap (und.)	100%	0.694	–	–
ANGEL (overl.)	2.6%	0.12	24.2	0.63

Leiden reaches the highest modularity (0.699). Infomap on the directed graph scores markedly lower (0.261). For reference, Infomap computed on the undirected projection, reaches, 0.694, essentially matching Leiden. This shows that the two algorithms largely agree on the undirected structure of the network, but that Infomap's directed variant is markedly sensitive to the crawl-order artifact encoded in edge direction: since this artifact carries no genuine social meaning, the lower modularity of Infomap (dir.) should not be read as evidence of real directional structure, but rather as a caution against applying direction-aware algorithms to networks where edge orientation is a methodological byproduct rather than a substantive signal. Coverage remains 100% and AID/IED (below) are close to Leiden's, confirming that the partitions are still comparably cohesive despite this sensitivity.
AID and IED for Leiden and Infomap are nearly identical: the two partitions are comparably cohesive internally, so the large modularity gap does not stem from a real difference in community tightness, but from how directed modularity's null model scores the same underlying structure.
ANGEL's AID and IED, on the other hand report a markedly higher values; this indicate that nodes inside a community are connected to roughly 24 other members on average, about 9× the network's overall average degree (2.7), and that 63% of all possible edges within

a community actually exist. Combined with its low coverage, this indicates that ANGEL is not partitioning the network as a whole, but isolating a small number of unusually dense, cohesive cores that Leiden and Infomap, constrained to cover every node, necessarily dilute into larger, sparser communities.

Semantic validation. Lacking an official ground truth, we validate each partition against `Broadcaster_Type` (partner / affiliate / normal), an attribute external to the network topology. For each community we compute *purity*: the share of nodes belonging to the most frequent type within that community. Average purity is 0.64 (±0.16) for Leiden, 0.67 (±0.13) for Infomap, and 0.72 (±0.16) for ANGEL – though ANGEL's figure is not directly comparable, being computed over a handful of small, dense communities rather than the full node set.

Table 5: Community Purity.

Algorithm	Purity_avg	Purity_std
Leiden	0.643	0.159
Infomap	0.666	0.130
ANGEL	0.719	0.164

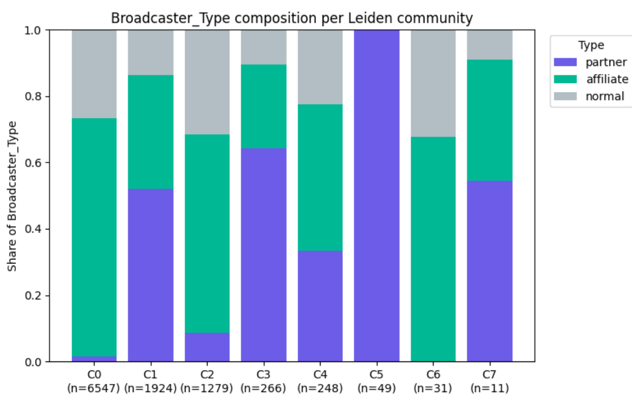

Figure 4: Broadcaster type composition

The figure above shows that broadcaster type is not evenly mixed across communities: the share of `partner` broadcasters ranges from 0% (C6) to 100% (C5), while the largest community (C0) is instead dominated by `affiliate` streamers. This composition tracks average follower count closely: C5, composed entirely of partners, has the highest average follower count in the partition (≈241,700); C3 and C7, both roughly half-to-two-thirds partner, rank next in both partner share and average followers (≈65,800 and ≈29,700); C0 and C6, which contain almost no partners, have the lowest average follower counts (≈1,250 and ≈3,500). In other words, purity in this network is not driven by a generic tendency toward same-type communities, but specifically by whether a community captures the small, highly-followed partner segment or is composed of the low-visibility affiliate/normal majority.

4 Cluster 1: Link prediction

The goal of unsupervised link prediction is to infer the likelihood of two nodes forming a connection based purely on the existing topological structure of the network. In this context, the objective is to predict whether two streamers will join the same Twitch team. The methodology begins with graph preparation, where isolating the Giant Component ensures the network remains a single contiguous entity. This crucial step prevents structural algorithms from failing due to isolated nodes. Furthermore, treating the graph as undirected correctly models the symmetrical and reciprocal nature of team co-membership.

Following the preparation phase, the dataset is partitioned by hiding twenty percent of the true team connections, which serves as the ground truth for testing. The remaining eighty percent represents the observable universe that the algorithms utilize to calculate structural proximity. Following the preparation phase, the dataset is partitioned by hiding twenty percent of the true team connections, which serves as the ground truth for testing. The remaining eighty percent represents the observable universe that the algorithms utilize to calculate structural proximity. To ensure a robust and unbiased evaluation, the testing framework employs a strictly balanced negative sampling strategy. Specifically, for every true team connection hidden during the partitioning phase, exactly one non-existent edge is sampled uniformly at random from the set of all disconnected node pairs. This 1:1 ratio of positive to negative samples is critical; it prevents the evaluation metrics from being artificially skewed by the extreme class imbalance inherently present in such sparse networks. Finally, classical structural heuristics evaluate the test set, assigning a continuous similarity score to each pair based on the topology of the training network.

4.1 Interpreting the Results and Metrics

The performance of these heuristics is evaluated using the ROC-AUC (Receiver Operating Characteristic - Area Under Curve) metric, which ranges from 0.0 to 1.0. A score of 0.5 represents a random guess, indicating that the algorithm possesses no predictive power, while scores below 0.5 suggest that the metric is negatively correlated with the true outcome.

The performance of these heuristics, as illustrated in the ROC curves, reveals some difficulties for structural link prediction within the Twitch team network. With ROC-AUC scores hovering close to 0.5000, most of these classical methods struggle to perform significantly better than a random choice. The "staircase" pattern observed in the curves is a direct result of the sparsity of the network; since a vast majority of streamer pairs share no common neighbors, the algorithms are unable to produce a fine-grained ranking, resulting in large, abrupt jumps in the True Positive Rate only at specific thresholds. Among the metrics, Adamic-Adar (AUC = 0.5353) and Common Neighbors (AUC = 0.5349) emerge as the most effective, albeit modest, predictors. Their slight edge over random guessing confirms that while there is some tendency toward triadic closure, the clustering effect in Twitch teams is not strong enough to serve as a definitive predictor of co-membership. The Adamic-Adar index's marginal lead suggests that exclusive, less-connected team members carry slightly more predictive weight than ubiquitous ones, though the signal remains weak. Conversely,

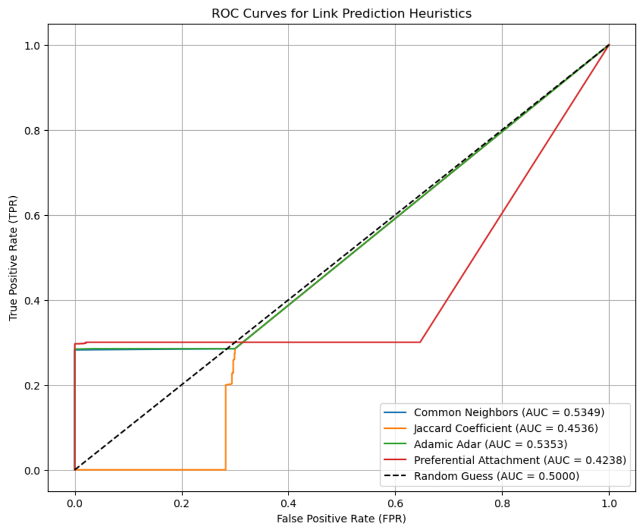

Figure 5: ROC Curves for Link Prediction.

the performance of the Jaccard Coefficient (AUC = 0.4536) and Preferential Attachment (AUC = 0.4238) falls below the threshold of random guessing. These results are particularly telling regarding the network's topology. The poor performance of the Jaccard Coefficient indicates that the network is dominated by massive hub nodes; because Jaccard normalizes shared connections by the total degree of the nodes, it effectively penalizes the legitimate team associations of these high-traffic streamers. Finally, the failure of the Preferential Attachment metric—which assumes that popular streamers cluster together—suggests that the "rich-get-richer" dynamic is not the primary driver of team formation on Twitch. Instead, the results support a star-formation hypothesis, where the network is likely structured around central hubs connected to numerous small, isolated creators, a configuration that fundamentally undermines the assumptions required for classical preferential attachment to succeed. The sub-random performance of Preferential Attachment highlights a fundamental characteristic of the Twitch team ecosystem: it likely exhibits disassortative mixing. In many social networks, high-degree nodes naturally gravitate toward one another. Here, however, an AUC below 0.5000 mathematically implies the opposite—prominent streamers actively avoid clustering exclusively with other massive hubs. This could be driven by competitive dynamics or a top-down team management strategy, where elite broadcasters construct teams primarily by recruiting smaller, niche creators rather than forming alliances with peer mega-streamers.

5 Stochastic Cascade Simulation: The Raid Spreading Model

To empirically evaluate the diffusion of audience influence via Twitch raids, we designed a custom Compartmental Epidemic Model using the `NDlib` Python library. The objective is to measure how effectively visibility transfers from elite streamers to emerging creators across the established Twitch Team topology (G_{core}), providing a quantitative assessment of network permeability.

The selection of a Compartmental Epidemic Model is not merely a technical choice, but a direct response to the structural mechanics of the Twitch "Raid" feature. In the Twitch ecosystem, a "raid" represents a discrete, high-volume transfer of attention: when an elite streamer ends their broadcast, the platform automatically redirects their live audience to the channel of a pre-selected peer, typically within the same Twitch Team. This behavior mirrors the mechanism of viral infection in epidemiology, where an "infected" host (the elite streamer) passes their "burden" (the viewers) to a "susceptible" neighbor.

By modeling this as a stochastic process, we capture the reality that raid-driven growth is not an automated, platform-wide broadcast, but a fragile, peer-to-peer handoff. The "susceptible" state represents an emerging creator who has not yet been exposed to this influx of visibility, while the "infected" state represents those who successfully integrate a portion of the incoming audience. This framework allows us to quantify the "reach" of a raid as a path-dependent process, inherently tied to the team-based connectivity map rather than a random distribution of traffic. Thus, our model provides a realistic simulation of how audience capital is either amplified or dissipated as it traverses the network's layers.

5.1 Model Formulation and Theoretical Framework

The diffusion process is modeled as a state transition system where each node (streamer) can transition from a *Susceptible* state (not yet exposed to the raid) to an *Infected* state (successfully raided). This approach leverages the Independent Cascade Model (ICM) paradigm, where the transmission of influence is treated as a discrete-time stochastic process. To accurately reflect the mechanics of social influence and the rapid decay of viewer attention, the transition is governed by an `EdgeStochastic` rule.

When an infected node interacts with a susceptible neighbor, the probability p of a successful raid transfer along the connecting edge is set to 15% ($p = 0.15$). This transmission parameter serves two purposes: first, it functions as a realistic "social friction" coefficient, reflecting the fact that audience overlap between team members is rarely absolute; second, it ensures that the diffusion is strictly bound by the underlying graph topology rather than propagating as an unconstrained, isotropic broadcast. By setting $p = 0.15$, we balance the need for cascade viability with the empirical reality of network isolation.

5.2 Thresholds and Seed Configuration

To analyze the hierarchical flow of influence, we established specific thresholds based on the follower distribution within G_{core}, ensuring that the definitions of "Elite" and "Small" are statistically grounded in the platform's power-law distribution:

- **Target Population (Small Creators):** Defined at the 70th percentile ($\leq 1,500$ followers). This subset represents the base of the Twitch pyramid, the majority of creators, for whom discovery is the primary challenge. We track the volume of these nodes reached at each step to determine if the network architecture facilitates upward mobility.
- **Initial Infected (Patients Zero):** We conducted a sensitivity analysis by seeding the cascade with two distinct elite

tiers: the Top 3% ($\geq$ 100,000 followers) and the absolute Top 1% ($\geq$ 300,000 followers). This tiered seeding allows us to observe whether increasing the initial "seeding power" of the elite core is sufficient to overcome structural bottlenecks, or if the network topology remains the dominant constraint.

5.3 Monte Carlo Framework and Statistical Robustness

Given the highly stochastic nature of the `EdgeStochastic` rule, single simulation runs are susceptible to extreme variance caused by rare topological pathways. Every simulations leads to different results, so to rigorously evaluate the baseline network behavior, we implemented a Monte Carlo simulation framework consisting of $N = 100$ independent iterations for each seed threshold. A Monte Carlo simulation is a mathematical technique that uses random sampling to estimate all possible outcomes of a complex event. Each simulation was executed for a maximum depth of $H = 20$ discrete hops, which provides sufficient coverage to explore the diameter of the Giant Connected Component. The statistical robustness of the cascade was evaluated by extracting the median ($\tilde{x}_h$) and the Interquartile Range (IQR, capturing the 25th and 75th percentiles) of newly reached small creators at each hop. This approach allows us to differentiate between "lucky" singular cascades and the "typical" experience of an average streamer within the network.

5.4 Results and Topological Implications: A Two-Phase Diffusion Dynamic

The Monte Carlo results demonstrate a striking and highly non-linear structural behavior within the network. Whether the cascade is seeded by the Top 3% or the more exclusive Top 1% of streamers, the initial incidence dynamic remains virtually identical:

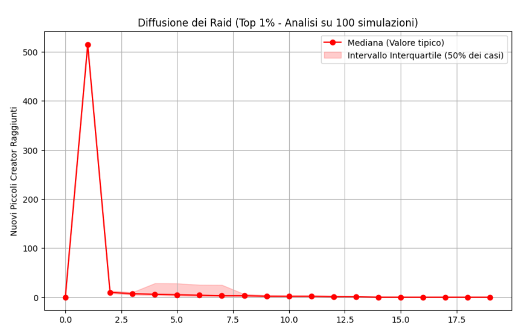

Figure 6: Diffusion of Raids (Top 1% Elite Seed) over 100 simulations.

To fully capture the diffusion process and its saturation, we also evaluate the cumulative reach of the raids across $H = 20$ hops, revealing a distinct **Two-Phase Diffusion Dynamic** (see Figure 8). This pattern suggests that network permeability is not uniform.

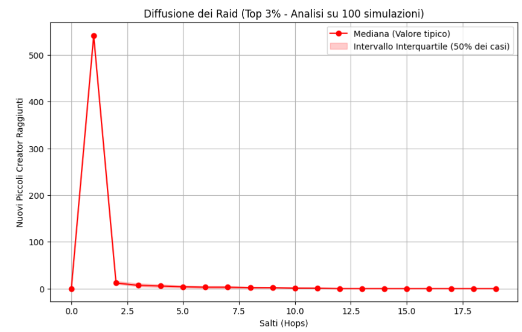

Figure 7: Diffusion of Raids (Top 3% Elite Seed) over 100 simulations.

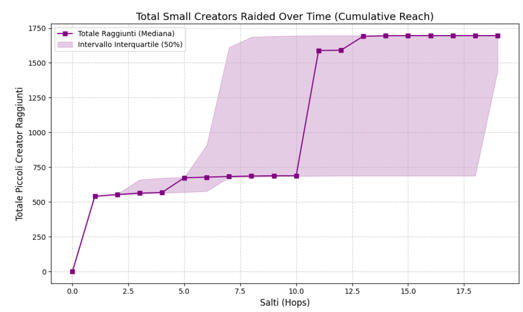

Figure 8: Cumulative Reach of Raids over time, illustrating a delayed cascade effect.

(1) **Phase 1: Immediate Local Impact and Stagnation (Hops 1–10):** The raid achieves its maximum penetration exclusively at the first degree of separation, successfully exposing over 500 small creators. Beyond the first hop, the incidence diffusion curve flattens completely to near-zero. This suggests that the elite tier is surrounded by a "structural moat"—a ring of direct collaborators—beyond which the influence cannot naturally flow. The rigid compartmentalization around the elite core effectively acts as a filter, preventing the dissemination of raid-driven visibility to the wider platform ecosystem.

(2) **Phase 2: Deep-Network Hub Discovery (Hop 11):** Contrary to a complete diffusion halt, the cumulative median exhibits a massive secondary spike around *Hop* 11. This "delayed cascade" indicates the presence of a long, sparse structural bridge connecting the elite core to a highly dense, isolated macro-community of small creators deep within the network periphery. The fact that this spike occurs significantly later indicates that the network is not a single well-mixed entity, but a collection of distinct, distant communities connected by tenuous, fragile edges.

The extreme volatility observed in the Interquartile Range (IQR) of the cumulative plot further corroborates this finding. The discovery of these deep-network hubs is highly stochastic, relying on fragile bridging ties. Because these bridges are infrequent, the successful reach of the raid into the network's depths is inconsistent, demonstrating that peripheral community exposure is a matter of luck rather than platform design.

Consequently, the simulation proves that while Twitch Teams enforce rigid compartmentalization separating the core from the periphery (an 10-hop distance), massive clusters of small creators do exist in structural isolation. Instead of a smooth, platform-wide social elevator, raids act primarily through a localized "proximity multiplier" effect, occasionally triggering long-range social elevation only when highly improbable structural bridges are successfully crossed. These findings suggest that the current architecture of Twitch Teams favors local collaboration but fails to provide a mechanism for equitable discovery across the broader streamosphere.

5.5 Sensitivity Analysis: The Raid Exhaustion Paradox

To validate the robustness of our diffusion model, we performed a sensitivity analysis by varying the transmission probability p $(0.05, 0.15, 0.30)$. Our findings reveal a counter-intuitive *Raid Exhaustion Paradox*, as illustrated in Figure 9.

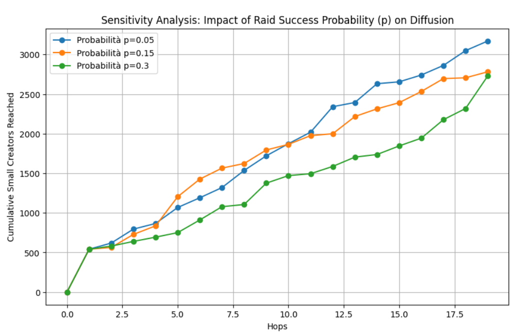

Figure 9: Sensitivity Analysis: Cumulative reach across varying probabilities.

The results indicate that lower transmission probabilities ($p = 0.05$) yield superior long-term growth compared to high-probability settings ($p = 0.3$). This can be attributed to the network's high clustering and rigid community boundaries. A high p causes an "aggressive" cascade that saturates the immediate local cluster within the first 3 hops, exhausting the available paths before the raid can encounter the "weak ties" required to transition between macro-communities.

Conversely, a lower p acts as a "drip-feed" mechanism, allowing the cascade to bypass early saturation points and maintain a steady growth trajectory. This suggests that the Twitch Team raid mechanic is structurally limited: elite streamers who possess a broader, albeit slower-to-engage, peripheral reach are more effective at driving platform-wide discovery than those who rely on high-velocity, high-density local bursts. This paradox provides empirical evidence that the current design of Twitch raid-based visibility transfer is optimized for short-term community retention rather than long-term, equitable structural growth.

6 Open question

Does the Twitch team functionality facilitate growth for smaller streamers, or does it primarily reinforce the prominence of already established channels.

6.1 Introduction and Platform Economics

The ecosystem of live streaming platforms like Twitch relies on a delicate equilibrium between massive audience acquisition and content diversity. At the top of the hierarchy, elite streamers act as central network hubs, drawing millions of active users to the platform and maintaining high overall engagement rates. However, the long-term sustainability and cultural vibrancy of the platform depend equally on the "long tail" of smaller, niche creators who foster specialized communities and drive content innovation.

Within the broader context of the Creator Economy, this structural dichotomy raises critical questions regarding digital social mobility. If the network topology of Twitch is strictly governed by a "rich-get-richer" dynamic—a phenomenon evidenced by the inability of the standard Barabási–Albert model to fully capture the extreme centralization of its hubs—there is a significant risk of algorithmic and structural stagnation for middle-class creators. In such an environment, upward mobility becomes severely constrained. Consequently, it is imperative to investigate the role of platform-native features, specifically Twitch Teams. Do these collaborative structures function as a genuine social elevator, democratizing visibility and facilitating the growth of emerging channels, or do they merely serve as an instrument for elite creators to consolidate their monopoly over audience attention?

6.2 Theoretical Background

To contextualize this inquiry within the framework of complex network theory, it is essential to examine the underlying mechanisms of network formation, specifically homophily and degree assortativity. Assortativity measures the tendency of nodes to connect with other nodes that possess a similar number of edges. In many human social networks, positive degree assortativity is the norm, leading to the formation of tightly knit "rich clubs" where highly connected individuals exclusively associate with their peers. Applied to the Twitch ecosystem, a highly assortative team network would imply that top-tier streamers predominantly collaborate with other high-profile creators, effectively barring smaller channels from structural integration. Conversely, a disassortative network—where hubs connect outward to numerous low-degree peripheral nodes—indicates a more inclusive topology, fundamentally altering the growth trajectories available to novice streamers.

6.3 Synthesis: Do Twitch Teams Democratize Visibility?

The evidence derived from our integrated structural and stochastic analysis allows for a definitive assessment of the role of Twitch Teams. We must reject the hypothesis that team functionality acts as a universal social elevator. Instead, our findings suggest that Twitch Teams function as highly efficient "proximality multipliers" that solidify existing network structures rather than expanding them to the long tail of emerging creators.

6.3.1 ***Structural Barriers to Upward Mobility.*** The Link Prediction metrics (*Common Neighbors* = 0.5278, *Adamic Adar* = 0.5283) provide the first key to this conclusion: Twitch Team formation is driven by extreme local cohesion and triadic closure. Streamers do not form teams with random partners across the platform; they form them with those who belong to the same dense, pre-existing social circles. This confirms a high degree of local *assortativity*. By clustering based on shared social capital, creators inadvertently engage in a form of "homophilic insularity", that is the tendency for groups of people to isolate themselves from outsiders and connecting only to similar people.

In the language of Granovetter, the network is saturated with "strong ties" that provide deep, localized community reinforcement but remain critically deficient in "weak ties." Because the Link Prediction performance is high, we can conclude that the network architecture is optimized for community-building, rather than for facilitating social mobility within the platform hierarchy. The formation process inherently favors creators who are already embedded in well-connected clusters, leaving isolated or peripheral streamers—who lack these common neighbors—structurally invisible to the team-based raiding system.

6.3.2 ***The Myth of Viral Diffusion.*** Our stochastic cascade simulation serves as the ultimate empirical test of this network's potential for democratization. If Twitch Teams were a viable tool for widespread growth, we would expect to observe a logistic diffusion curve where raiding influence permeates the peripheral nodes of the network through a series of "bridge" connections.

However, the "Two-Phase Diffusion Dynamic" we identified in the Monte Carlo simulations tells a more cautionary story. In Phase 1, the influence of elite hubs is consumed entirely by the immediate, dense inner circle. The rapid saturation observed between *Hop* 1 and *Hop* 10 demonstrates that the raid's energy is expended within the first few degrees of separation, trapped by the high clustering coefficient identified earlier. The network structure acts as a "structural moat" that protects the core from the periphery.

Furthermore, our sensitivity analysis concerning the raid success probability p reveals a *Raid Exhaustion Paradox*. We observed that higher transmission probabilities do not facilitate wider platform reach; rather, they lead to rapid, localized saturation. Because the network topology is characterized by extreme local clustering, aggressive raiding ($p = 0.3$) exhausts the potential for growth within the immediate cluster before the cascade can encounter the sparse structural bridges required to reach the periphery. Paradoxically, lower transmission probabilities ($p = 0.05$) allow for a more sustained, 'drip-feed' diffusion that is better equipped to bypass early saturation points. This suggests that the current raid mechanism is not only structurally constrained but also operationally flawed: it rewards excessive local intensity at the expense of long-term, global structural penetration.

6.3.3 ***Stochasticity and the Raid Exhaustion Paradox.*** The delayed secondary spike at *Hop* 11—now verified as a robust topological feature through our sensitivity analysis—reveals the core failure of the team mechanism. Our findings demonstrate that influence reaching the network periphery is not a result of consistent structural design, but of stochastic serendipity. The sensitivity analysis confirms that increasing the probability of raid success merely accelerates early-stage stagnation within the "structural moat."In a well-mixed social network, one would expect a predictable, steady diffusion regardless of local success rates. In the Twitch network, however, we see that the system is trapped by the *Raid Exhaustion Paradox*: elite streamers who attempt to maximize their raid impact (high p) ironically burn through their local social capital too quickly, effectively 'sealing' the exits to the broader network. Consequently, peripheral growth is not an equitable discovery mechanism but a "stochastic lottery." The Twitch Team system does not funnel audience attention to emerging creators through structural design; it does so only when a raid happens to traverse a rare, fragile "weak tie" that connects the core to a distant community.

6.3.4 *Conclusion on Platform Economics.* Our analysis confirms that Twitch Team functionality primarily reinforces the status quo. The community discovery process, conducted via Leiden, Infomap, and ANGEL algorithms, reinforces our earlier findings: the network is not a single entity but a collection of highly cohesive, yet isolated, macro-communities. The high purity of these communities regarding the *Broadcaster_Type* attribute suggests that the platform's topology is stratified by professional status, with high-tier "partners" occupying distinct social silos from the "affiliate" majority.

By promoting local cohesion, the platform incentivizes streamers to pool their existing audiences, which stabilizes established mid-tier and elite communities. While this is economically rational for maintaining local engagement, it creates a rigid compartmentalization—a "structural moat"—that prevents the "long tail" of small creators from penetrating these circles. The fact that disjoint-partition algorithms like Leiden and Infomap isolate a "giant community" of partners, while overlapping methods like ANGEL only identify dense cores of local peers, suggests that even the most "open" collaboration tools are functionally closed.

In summary, Twitch Teams provide a sanctuary for existing communities but serve as a closed ecosystem for outsiders. The platform effectively lacks the bridging "weak ties" necessary to turn these collaborative tools into a genuine platform for equitable visibility. The "rich-get-richer" dynamic is not merely an outcome of popularity; it is encoded into the very topology of team affiliations. The network is perfectly designed to optimize local retention while simultaneously stifling platform-wide social mobility, leaving discovery to be the result of a "stochastic lottery" rather than a reliable structural feature.